

Math 548
Portfolio Assignment 10

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Problem 1 (F20, P4). Let R be a commutative ring with 1. We say an element $n \in R$ is **nilpotent** if there exists a number $k \in \mathbf{N}$ such that $n^k = 0$.

- (a) Show that if n is nilpotent, then $1 - n$ is a unit.
- (b) Give an example of a commutative ring with 1 that has no nonzero nilpotent elements, but is not an integral domain.

Proof. (a) Since n is nilpotent, there exists $k \in \mathbf{N}$ such that $n^k = 0$. Note that $\frac{1-n^k}{1-n} = \sum_{i=0}^{k-1} n^i$, hence $\left(\sum_{i=0}^{k-1} n^i\right)(1-n) = (1-n^k) = 1$. Thus $1 - n$ is a unit.

(b) Consider the ring $\mathbf{Z}/6\mathbf{Z}$. Note that this space is not an integral domain as $[3]_6[2]_6 = [0]_6$. Suppose $[a]_6^n = [0]_6$. Then $6 \mid a^n$. It follows that $2 \mid a^n$ and $3 \mid a^n$. Since 2 and 3 are prime, we have that $2 \mid a$ and $3 \mid a$. Since 2 and 3 are coprime, we have $6 \mid a$. Hence $[a]_6 = [0]_6$. Thus $\mathbf{Z}/6\mathbf{Z}$ has no nonzero nilpotent elements. $\square$

Problem 2 (F20, P5). Let R be a ring with 1 and suppose $e \in R$ is idempotent; i.e. satisfies $e^2 = e$.

- (a) Prove that $1 - e$ is also idempotent.
- (b) Suppose $e \neq 0, 1$. Show that Re and $R(1 - e)$ are proper left ideals of R .
- (c) Prove there is an isomorphism $R \cong Re \times R(1 - e)$.

Proof. (a) We have:

$$\begin{aligned}(1 - e)^2 &= 1 - 2e + e^2 \\ &= 1 - 2e + e \\ &= 1 - e.\end{aligned}$$

Thus $1 - e$ is idempotent.

(b) Note that Re and $R(1 - e)$ are nonempty. Let $r_1, r_2 \in R$. We have:

$$\begin{aligned}r_1e - r_2e &= (r_1 - r_2)e \in Re, \\ r_1(1 - e) - r_2(1 - e) &= (r_1 - r_2)(1 - e) \in R(1 - e).\end{aligned}$$

Let $re \in Re$, $r(1 - e) \in R(1 - e)$, and $s \in R$. We have:

$$\begin{aligned}s(re) &= (sr)e \in Re \\ s(r(1 - e)) &= (sr)(1 - e) \in R(1 - e).\end{aligned}$$

Thus Re and $R(1 - e)$ are proper ideals.

(c) Define $\varphi : R \rightarrow Re \times R(1 - e)$ by $r \mapsto (re, r(1 - e))$. Let $r, s \in R$. Observe that:

$$\begin{aligned}\varphi(rs) &= (rse, rs(1 - e)) \\ &= (rse^2, rs(1 - e)^2) \\ &= (re, r(1 - e))(se, s(1 - e)) \\ &= \varphi(r)\varphi(s).\end{aligned}$$

$$\begin{aligned}\varphi(r + s) &= ((r + s)e, (r + s)(1 - e)) \\ &= (re + se, r(1 - e) + s(1 - e)) \\ &= (re, r(1 - e)) + (se, s(1 - e)) \\ &= \varphi(r) + \varphi(s).\end{aligned}$$

Now suppose $\varphi(r) = \varphi(s)$. Then $(re, r(1 - e)) = (se, s(1 - e))$, hence $re = se$. This is equivalent to $(r - s)e = 0$. But since $e \neq 0$, it must be the case that $r - s = 0$. Hence $r = s$. Thus φ is injective.

Let $(x, y) \in Re \times R(1 - e)$ be any elements. Then $x = re$ and $y = s(1 - e)$ for some $r, s \in R$. We have:

$$\begin{aligned}\varphi(re + s(1 - e)) &= ((re + s(1 - e))e, (re + s(1 - e))(1 - e)) \\ &= (re^2 + se - se^2, re + s - e - re^2 - se - e^2) \\ &= (re, s - se) \\ &= (re, s(1 - e)).\end{aligned}$$

Thus φ is a ring isomorphism. □

Problem 3 (S16, P3). Let R be a commutative ring.

- (a) Prove that the set N of all nilpotent elements of R is an ideal.
- (b) Prove that R/N is a ring with no nonzero nilpotent elements.
- (c) Show that N is contained in every prime ideal of R .

Proof. (a) Note that N is nonempty since $0 \in N$. Moreover, if $x \in N$, then clearly $-x \in N$ as $(-x)^n = (-1)^n x^n$. Let $r_1, r_2 \in N$. Then $r_1^n = 0$ and $r_2^k = 0$ for some $n, k \in \mathbf{N}$. Observe that:

$$\begin{aligned}(r_1 + r_2)^{n+k} &= \sum_{i=0}^{n+k} \binom{n+k}{i} r_1^{n+k-i} r_2^i \\ &= \sum_{i=0}^{n+k} \binom{n+k}{i} r_1^n r_1^k r_1^{-i} r_2^i \\ &= 0.\end{aligned}$$

Thus $r_1 + r_2 \in N$, hence N is closed under addition. Now let $x \in N$ and $a \in R$. Then there exists $n \in \mathbf{N}$ such that $x^n = 0$. We can see $(ax)^n = a^n x^n = 0$, hence $ax \in N$. Thus N is an ideal.

(b) Suppose $[x] \in R/N$ is nilpotent. Then there exists $k \in \mathbf{N}$ such that $[x]^k = [x^k] = [0]$. This means $x^k \in N$, so there exists $n \in \mathbf{N}$ such that $(x^k)^n = x^{kn} = 0$. But this means $x \in N$, hence $[x] = [0]$.

(c) Let $r \in N$. Then $r^n = 0$ for some $n \in \mathbf{N}$. Since every prime ideal contains 0, every prime ideal contains r^n . Hence every prime ideal contains r . Thus N is contained in every prime ideal of R . $\square$

Problem 4 (S12, P4). Let R be a commutative ring with 1 and let:

$$N = \{r \in R \mid r^n = 0 \text{ for some } n\}$$

be the *nilradical* of R (i.e., the set of all nilpotent elements). You may assume N is an ideal. Prove the following are equivalent:

- (1) Every element of R is either nilpotent or a unit.
- (2) R/N is a field.

Proof. ($\Rightarrow$) Let $[r] \in R/N$ be nonzero. Then r is not nilpotent, hence r must be a unit. So there exists $s \in R$ such that $rs = 1$. This means $[r][s] = [rs] = [1]$. Since every $[r]$ was arbitrary, every element of R/N has a unit; i.e., R/N is a field.

($\Leftarrow$) Let $a \in R$ not be a nilpotent element. Then $[a] \in R/N$. Since R/N is a field, there exists $[b] \in R/N$ such that $[a][b] = [1]$. Hence $ab - 1 \in N$. This means there exists some $n \in \mathbf{N}$ such that $(ab - 1)^n = 0$. Hence $\sum_{k=0}^n \binom{n}{k} ab^k (-1)^{n-k} = 0$. We can rewrite this as:

$$(-1)^n + \sum_{k=1}^n \binom{n}{k} (ab)^k (-1)^{n-k} = 0$$

Without loss of generality, suppose that n is odd (if it were even, then multiply both sides by the following equation by -1). Then we have:

$$\begin{aligned} 1 &= \sum_{k=1}^n \binom{n}{k} (ab)^k (-1)^{n-k} \\ &= a \sum_{k=1}^n \binom{n}{k} a^{k-1} b^k (-1)^{n-k}. \end{aligned}$$

Hence a is a unit. $\square$

Problem 5 (Midterm 2). Let D be a principal ideal domain. Prove that every proper nonzero prime ideal is maximal.

Proof. Let $I \subset D$ be a proper nonzero prime ideal. Then $I = (p)$ for some prime p . Suppose there is an ideal $(m) \subseteq D$ such that $(p) \subseteq (m)$. Then $p = sm$ for some $s \in \mathbf{Z}$. Since (p) is prime and $sm \in (p)$, either $m \in (p)$ or $s \in (p)$. If $m \in (p)$, then $(p) = (m)$, hence (p) is maximal. If $s \in (p)$, then $s = pt$ for some $t \in \mathbf{Z}$. In this case $p = ptm$, which is equivalent to $p(1 - tm) = 0$. Since D is an integral domain, $tm = 1$. Hence m is a unit, which means $(m) = R$. Thus (p) is maximal. $\square$

Problem 6. Suppose R is a ring such that $r^2 = r$ for every element $r \in R$.

- (a) Prove $r = -r$ for every element $r \in R$.
- (b) Show R must be commutative. Hint: consider $(a + b)^2$.

Proof. (a) Observe that:

$$\begin{aligned} r + r &= (r + r)^2 \\ &= r^2 + 2r^2 + r^2 \\ &= r + r + r + r. \end{aligned}$$

Hence $r + r = 0$. Thus $r = -r$.

(b) Observe that:

$$\begin{aligned} a + b &= (a + b)^2 \\ &= a^2 + ab + ba + b^2 \\ &= a + ab + ba + b. \end{aligned}$$

Hence $ab + ba = 0$. This means $ab = -ba$, and by part (a) $ab = ba$. Thus R is commutative. $\square$

Problem 7. Let A be a commutative ring with unit. We call A *Boolean* if $a^2 = a$ for every $a \in A$. Prove that in a Boolean ring A each of the following holds:

- (a) $2a = 0$ for every $a \in A$.
- (b) If I is a prime ideal then A/I is a field with two elements (and in particular I is maximal).
- (c) If $I = (a, b)$ is the ideal generated by a and b then I can be generated by the single element $a + b + ab$. Conclude that every finitely generated ideal is principal.

Proof. (a) Observe that:

$$\begin{aligned} 2a &= a + a \\ &= (a + a)^2 \\ &= a^2 + 2a^2 + a^2 \\ &= 4a. \end{aligned}$$

Hence $2a = 0$.

(b) Let $[x] \in A/I$ be nonzero. Then $[x]^2 = [x^2] = [x]$, hence $[x(x - 1)] = [0]$. This means $x(x - 1) \in \mathfrak{p}$, and since $\mathfrak{p}$ is a prime ideal either $x \in \mathfrak{p}$ or $(x - 1) \in \mathfrak{p}$. Since $x \neq 0$, $x \notin \mathfrak{p}$, so it must be the case that $(x - 1) \in \mathfrak{p}$. Hence $[x - 1] = [0]$; i.e., $[x] = [1]$. Thus A/I is a field with two elements.

(c) Clearly $(a + b + ab) \subseteq (a, b)$. For the converse direction, observe that:

$$\begin{aligned} a &= a + 2(ab) \\ &= a + ab + ab \\ &= a^2 + ab + a^2b \\ &= a(a + b + ab). \end{aligned}$$

$$\begin{aligned}
 b &= b + 2(ab) \\
 &= b + ab + ab \\
 &= ab + b^2 + ab^2 \\
 &= b(a + b + ab).
 \end{aligned}$$

Since $a, b \in (a + b + ab)$, we have $(a, b) \subseteq (a + b + ab)$. Thus $(a, b) = (a + b + ab)$; i.e., every finitely generated ideal of A is principal. $\square$